Algorithmic Trading with MARL in Limit Order Books - Key Answers

1. Data in the Limit Order Book

We rely primarily on simulation using ABIDES, an agent-based market simulator. In this setup, the limit order book is generated dynamically by interacting agents placing buy and sell orders. This approach allows full control of the environment and enables realistic multi-agent interaction.

Simulation is fully free, scalable, and suitable for reinforcement learning. However, it may lack realism and requires calibration to match real market characteristics such as spread, volatility, and order flow.

As an optional extension, free historical data (e.g., from crypto exchanges) can be used for validation or calibration. However, such data is limited and does not allow agents to influence the market, making it less suitable for training.

2. Training Method

We train our agents using reinforcement learning, not supervised learning. ABIDES acts as an environment in which agents interact with the market.

At each step, the agent observes the state of the order book, takes an action (placing or cancelling orders), and receives a reward based on its performance. The reward typically includes profit and penalties for risk or inventory.

This process allows the agent to learn optimal strategies through trial and error rather than relying on labeled data.

3. Evaluation Metrics

We evaluate performance using two key metrics. First, the Sharpe Ratio, which measures risk-adjusted returns by comparing average profit to its variability. This provides a standard benchmark for financial performance.

Second, we use inventory risk or maximum drawdown. These metrics capture the risk associated with holding positions and potential losses during adverse market movements, which is critical in market-making strategies.

4. Baseline Strategy

As a benchmark, we use a naive market-making strategy. This strategy continuously places a buy order below the mid-price and a sell order above it, earning the spread without adapting to market conditions.

This baseline is simple but realistic, and it captures the key trade-off between profitability and inventory risk. Our MARL agents are expected to outperform this benchmark by dynamically adjusting their behavior.